

IPv4 Addressing

1. Convert the following IP address to its 32-bit binary representation: 172.16.5.10. Write out each octet in binary, separated by dots.

->The IPv4 address **172.16.5.10** converted to 32-bit binary is:

- **172** → 10101100
- **16** → 00010000
- **5** → 00000101
- **10** → 00001010

Answer

10101100.00010000.00000101.00001010

2. Given the binary IP address 11000000.10101000.00000001.00001111, convert it back to standard dotted decimal notation.

->Convert each 8-bit binary octet to decimal:

- 11000000 → **192**
- 10101000 → **168**
- 00000001 → **1**
- 00001111 → **15**

Answer

192.168.1.15

3. You are given a list of 10 IP addresses used by different apps (e.g., 192.168.43.1, 8.8.8.8, 10.0.5.25, 172.20.14.2, 203.0.113.5, 172.15.255.255, 192.0.2.10, 192.168.1.100, 123.45.67.89, 172.16.0.1). For each, identify if it is a private or public IP address and explain your reasoning.

->Here's the classification using the standard IPv4 private ranges:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

IP Address	Type	Reason
192.168.43.1	Private	Falls within 192.168.0.0 – 192.168.255.255.
8.8.8.8	Public	Not within any private IP range; it is publicly routable.
10.0.5.25	Private	Falls within 10.0.0.0 – 10.255.255.255.
172.20.14.2	Private	Falls within 172.16.0.0 – 172.31.255.255.
203.0.113.5	Public (special-use)	Not a private IP range. It is reserved for documentation/examples, so it is not normally used as a real public address.
172.15.255.255	Public	172.15.x.x is outside the private range, which starts at 172.16.0.0.
192.0.2.10	Public (special-use)	Not a private range; 192.0.2.0/24 is reserved for documentation/examples.
192.168.1.100	Private	Falls within 192.168.0.0 – 192.168.255.255.
123.45.67.89	Public	Not within any RFC 1918 private range.
172.16.0.1	Private	It is the first address in the private 172.16.0.0 – 172.31.255.255 range.

Final classification

Private IPs:

192.168.43.1, 10.0.5.25, 172.20.14.2, 192.168.1.100, 172.16.0.1

Public/non-private IPs:

8.8.8.8, 203.0.113.5, 172.15.255.255, 192.0.2.10, 123.45.67.89

Important: 203.0.113.5 and 192.0.2.10 are technically **not private**; they are special addresses reserved for documentation and examples.

4. For the CIDR notations /20, /25, and /30, calculate the total number of usable IP addresses in each subnet. Show your working for each calculation.

Hint:
Usable addresses = ($2^{\text{number of host bits}}$) minus 2 (for network and broadcast addresses).

->**CIDR Subnet Usable IP Address Calculation**

The formula is:

$$\text{Usable IP addresses} = 2^{(\text{host bits})} - 2$$

IPv4 addresses have **32 bits**, so:

$$\text{Host bits} = 32 - \text{CIDR prefix}$$

1. /20

- Host bits = $32 - 20 = 12$
- Total addresses = $2^{12} = 4,096$
- Usable addresses = $4,096 - 2 = 4,094$

✓ **Answer: 4,094 usable IP addresses**

2. /25

- Host bits = $32 - 25 = 7$
- Total addresses = $2^7 = 128$
- Usable addresses = $128 - 2 = 126$

✓ **Answer: 126 usable IP addresses**

3. /30

- Host bits = $32 - 30 = 2$
- Total addresses = $2^2 = 4$
- Usable addresses = $4 - 2 = 2$

✓ **Answer: 2 usable IP addresses**

CIDR	Host Bits	Total Addresses	Usable Addresses
/20	12	4,096	4,094

CIDR	Host Bits	Total Addresses	Usable Addresses
/25	7	128	126
/30	2	4	2

5. Imagine you are setting up a small LAN for a gaming night with friends using your WiFi router at home. Assign static private IP addresses to 5 devices (laptop, phone, tablet, smart TV, and PlayStation) using the 192.168.0.0/24 range. List the IP you would assign to each device and explain why you chose that range.

->Static IP Address Plan for a Small Gaming LAN

For the gaming night, I would use the **192.168.0.0/24** private network. This provides addresses from **192.168.0.1** to **192.168.0.254** for devices.

Device	Static IP Address	Reason
 Laptop	192.168.0.10	Easy to remember and leaves room for other devices.
 Phone	192.168.0.11	Kept close to the laptop for simple organization.
 Tablet	192.168.0.12	Sequential address makes the network easier to manage.
 Smart TV	192.168.0.13	A separate, predictable address for the TV.
 PlayStation	192.168.0.14	A fixed address makes it easy to identify on the LAN.

Why use 192.168.0.0/24?

The **192.168.0.0/24** range is a standard **private IPv4 network** commonly used for home and small-office LANs. The /24 means the first 24 bits identify the network, leaving **8 host bits**, giving 256 total addresses (192.168.0.0–192.168.0.255).

- 192.168.0.0 → Network address
- 192.168.0.1–192.168.0.254 → Usable device addresses
- 192.168.0.255 → Broadcast address

Using static addresses such as **.10–.14** makes the five gaming-night devices easy to identify and manage.

